Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

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Reminder and plan for today

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

Design

Staggered difference-in-differences with event study

Treatment: the UV, the year it opens its first public preschool

Control: UVs that never get one

Panel: UV $\times$ year, with UV and year fixed effects; woman fixed effects as a check

Effect: intent-to-treat of access, not individual take-up

Today

Sample changes: who enters and leaves, and why

Data fixes: recap and the UV crosswalk

Results, one by one: first stage, formal employment, earnings, fertility

What is pending from our conversations

What changed in the samples

Labor market

Who: vulnerable, with a child <5 in at least one year, 2014–2024

Enters: first year in the data, even before her first child
Before: first year with a child <5

Leaves: not because of age, within the 18–70 panel
Before: years after 49 without a child <5 were dropped

New: balanced version, all 11 years

Fertility

Who: vulnerable, aged 18–35 in 2014, with or without children
Before: 18–49 each year

Enters: first year in the data

Leaves: never because of age

New: balanced version, all 11 years

Why

Keep changes in who is observed out of the effects

Bring the panel as close to balanced as possible

Compare women at the same stage of motherhood

Women 18–35 in 2014 have 80% of births, and no one enters or leaves by turning 18 or 50

$\Rightarrow$ **We follow the same women for longer, so composition weighs less on the estimates.**

Who is in each sample

First stage

Children aged 0–4 of vulnerable mothers

Only years at those ages, if the mother is observed that year

Labor market

Defined by **being a mother**, at any age

817k mothers

Fertility

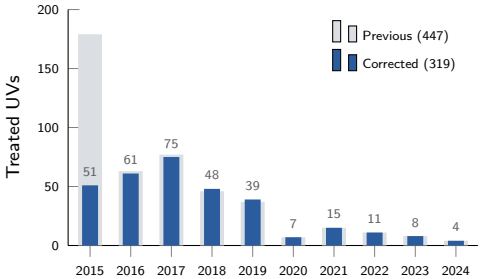
Defined by **age**, with or without children

1.02M women

In every sample: women aged 18–70 in the FPS/RSH, 2014–2024; vulnerable = 60% most vulnerable, fixed the first year she is classified. A woman has data only in years she is in the registry with a valid UV. Years 2014–2015 count only if she is also in 2016, and use her 2016 UV. Only UVs without a preschool in 2014: 319 treated, 4,174 never treated.

⇒ **The labor sample follows mothers; the fertility sample follows women of a fixed age group.**

Data fixes: recap



Opening years

Most 2015 "openings" were centers already operating

Treated UVs: **447** $\rightarrow$ **319**

Earnings

Before: only taxable (dependent) income if she had any

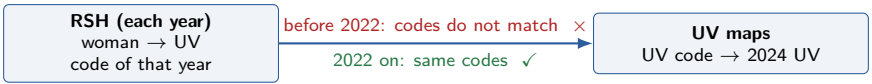
Now: taxable + independent income

UV crosswalk

Fixed again this week (next slide)

$\Rightarrow$ **These fixes correct who is treated and how earnings are measured.**

UV crosswalk: one rule for all years



So before 2022, we follow where each code's households live the next year

Before: UV codes treated differently by quality → the problem spread to more years

Now: every code follows the same rule, for all years

	Before	Now
Women changing UV per year	19–26%	**4–9%**
Of those, with the same RSH code	50–75%	**0–1%**
Treated UVs with data from 2014	241	**299**
First stage, $k = -2$	+0.9pp	**+0.3pp**

⇒ **The crosswalk improved: spurious moves are gone and the pre-trend in the first stage is smaller.**

UV crosswalk: the 2022 jump remains

What we see

Women changing UV in 2022: **23%**, vs. 4–9% other years

Cartography and method change that year

Sample UVs added in 2022: **203 control, 12 treated**, vs. 0–30 other years

Could mean 15–18% of women have another UV in 2016–2021 (inferred, not measured)

So far

Similar in both groups: **25.9% treated, 26.3% control**

Raw levels: **the treated − control gap does not jump in 2022**

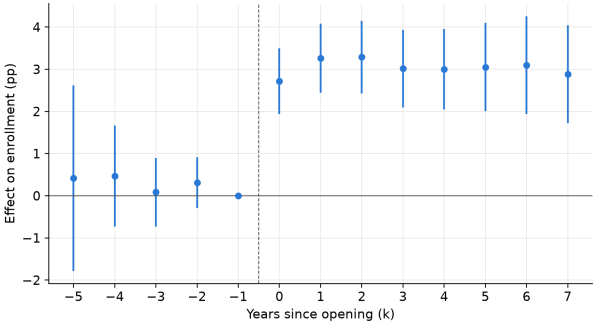
Next

Woman FE with the UV fixed from 2022

Without 2021 codes split across UVs

⇒ **In raw levels, results do not seem driven by 2022, but this jump needs a deeper look.**

First stage: enrollment rises the year a preschool opens



Measure: share of children 0–4 of vulnerable mothers enrolled

Year before opening: treated 31.7%, control 33.0%

Pre-trends: none ($p = 0.81$)

Estimate, first 8 years: **+3.0pp**, significant, ≈3.6 more children per UV

Last meeting: +3.6pp, with a significant $k = -2$ that is now gone

⇒ **Enrollment jumps when the first preschool opens and stays higher: from 34% to 37%, about 3.6 more children per UV.**

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$. Levels: control mean.

Labor market results: who we measure

Vulnerable mothers: a child <5 in at least one year, 2014–2024, at any age.

Full sample: **817k mothers**

Counts in the UV where she lives that year

Not counted in years in a UV with a preschool in 2014

Balanced: **247k mothers**

Same, only women with all 11 years in these UVs

Woman FE: **5.6M mother-years**

Compared with herself, in the first valid UV where she appears, even if she moves

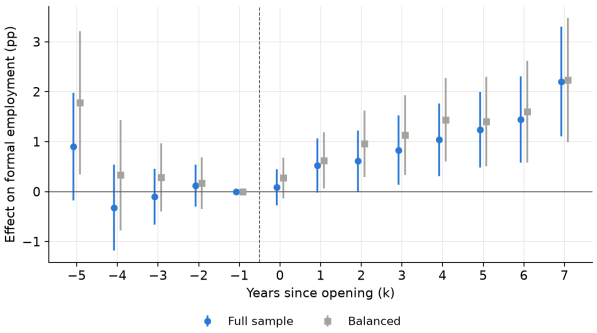
Treated UV: she must appear before the opening.
Control UV: any year

Woman FE, balanced: **357k mothers**

Same, only women who appear in 2014 and have all 11 years, in any UV

⇒ **The same mothers, measured four ways: from the UV where they live to each mother against herself.**

Employment: more mothers enter formal work, gradually



Measure: share of mothers with formal earnings in the year

Year before opening: 54.8% in both groups

Pre-trends: none ($p = 0.15$, balanced $p = 0.18$)

Estimate, first 8 years: **+1.0pp** full, **+1.2pp** balanced, both significant

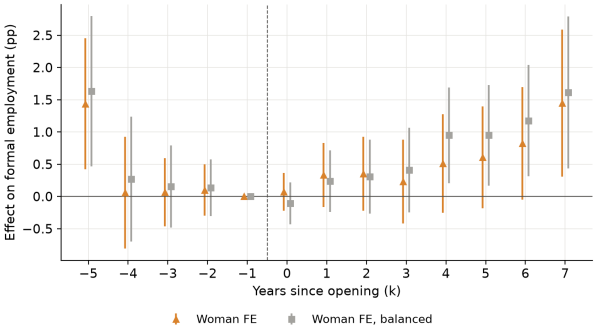
≈2.5 more working mothers per UV

Last meeting: +1.5pp, with negative pre-trend coefficients that are now around zero

⇒ When the first preschool opens, formal employment rises gradually: from 54.6% to 55.6% of mothers.

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$. Levels: control mean.

Employment, woman fixed effects: also rises gradually



Measure: whether each mother has formal earnings in the year

Woman fixed effects: each mother compared with herself, assigned to the first UV where she appears

Start: ≈55% of these mothers have formal earnings

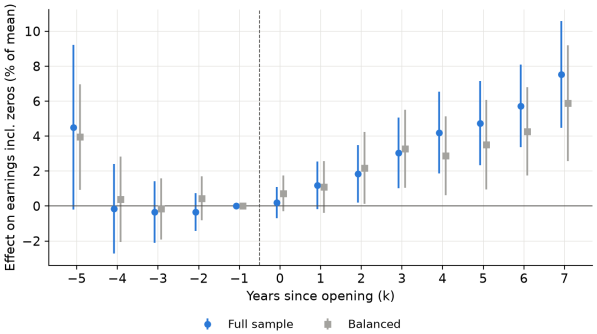
Pre-trends: none ($p = 0.06$, balanced $p = 0.10$)

Estimate, first 8 years: their share rises **+0.55pp**, **+0.7pp** balanced (significant)

⇒ **Following the same women, the share of mothers with formal earnings also rises gradually, at about half the size: from 54.7% to 55.3%.**

Sun–Abraham, 95% CI, ref. $k = -1$. Levels: control mean.

Earnings including zeros rise gradually



Measure: formal earnings of all mothers, zero if none, relative to the year's mean

Level, 2024 pesos: ≈CLP 260k a month in 2014, ≈CLP 424k in 2024; about half have no formal earnings

Pre-trends: none in full ($p = 0.13$), yes in balanced ($p = 0.04$)

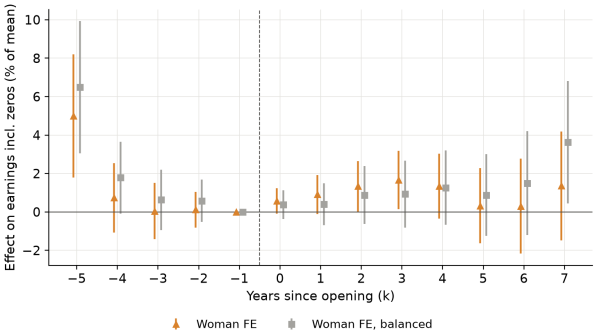
Estimate, first 8 years: **+3.6%** full, **+3.0%** balanced, both significant

Last meeting: +4.4%

⇒ **Mothers' average earnings in the UV rise gradually: from about CLP 335k to 347k a month, in 2024 pesos.**

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$. Levels: mean 2016–2024, approximate.

Earnings including zeros, woman fixed effects



Woman fixed effects: each mother compared with herself, in the first UV where she appears

Level, 2024 pesos: ≈CLP 260k a month in 2014, ≈CLP 424k in 2024

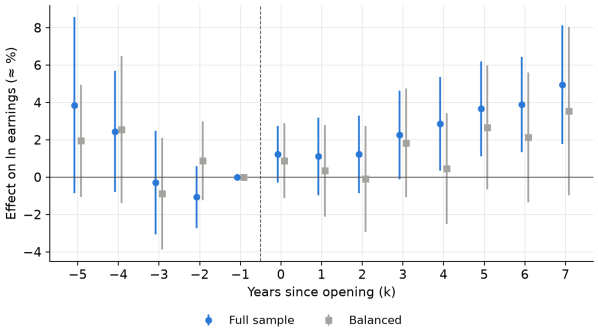
Pre-trends: yes ($p = 0.02$, balanced $p < 0.01$)

Estimate, first 8 years: **+1.0%**, **+1.2%** balanced, not significant

⇒ **With woman fixed effects, the rise is small and not significant: about CLP 3k a month, smaller than in the UV.**

Sun–Abraham, 95% CI, ref. $k = -1$. Levels: mean 2016–2024, approximate.

Earnings of mothers with earnings



Measure: ln annual earnings, only mothers with formal earnings that year

Level, 2024 pesos: ≈CLP 470k a month in 2014, ≈CLP 770k in 2024

Pre-trends: none ($p = 0.20$, balanced $p = 0.20$)

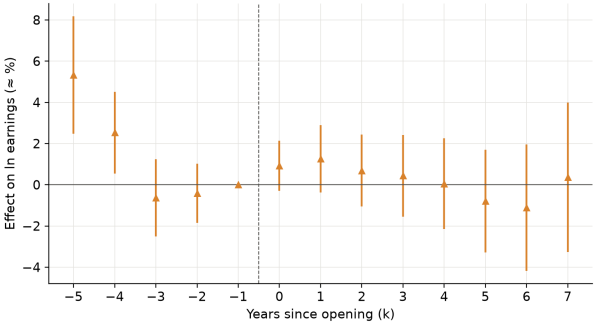
Estimate, first 8 years: **+2.65%** full, significant; **+1.5%** balanced, not significant

Annual: mixes wages and months worked; who has earnings can change

⇒ **Earnings of mothers with earnings rise about CLP 17k a month in the UV.**

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$. Levels: mean 2016–2024, approximate.

Earnings of mothers with earnings, woman fixed effects



Woman fixed effects: each mother compared with herself, only years with formal earnings

Level, 2024 pesos: ≈CLP 470k a month in 2014, ≈CLP 770k in 2024

Pre-trends: yes ($p < 0.01$)

Estimate, first 8 years: **+0.2%**, not significant

Women who start working after the opening have no earlier year to compare, so their entry into work is not captured

⇒ **With woman fixed effects we do not see earnings rise; with these pre-trends, it is the least reliable version.**

Sun–Abraham, 95% CI, ref. $k = -1$.

How much of the earnings rise comes from more mothers working?

Average earnings can rise because **more mothers work** or because **those who work earn more**

If only more mothers worked, each earning the mean (CLP 630k): +1pp is 1 more mother in 100, so the average rises by **CLP 6.3k**

First 8 years, 2024 pesos	More employment	Rise from employment alone	Actual rise	Share explained
Full sample	+1.0pp	CLP 6.3k	CLP 12k	≈half
Balanced	+1.2pp	CLP 7.6k	CLP 10k	≈three quarters
Woman FE	+0.55pp	CLP 3.5k	CLP 3.2k	≈all
Woman FE, balanced	+0.7pp	CLP 4.3k	CLP 4.1k	≈all

⇒ **More mothers working explains about half of the rise in the UV, and about all of it with woman fixed effects. The rest in the UV would come from mothers already working or from who lives there.**

Approximate: new workers assumed to earn mean earnings.

Fertility results: who we measure

Vulnerable women aged 18–35 in 2014, with or without children.

Full sample: **1.02M women**

Counts in the UV where she lives that year

Not counted in years in a UV with a preschool in 2014

Woman FE: **6.9M woman-years**

Compared with herself, in the first valid UV where she appears, even if she moves

Treated UV: she must appear before the opening.
Control UV: any year

Balanced: **302k women**

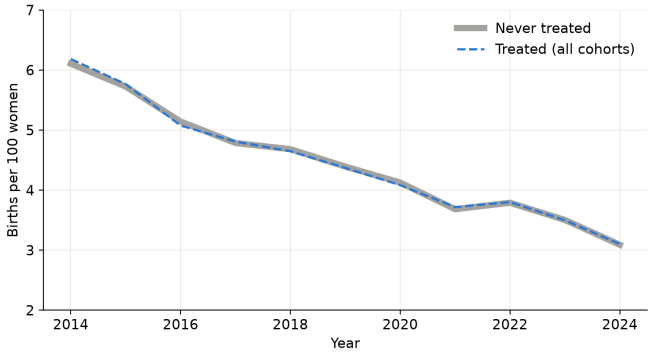
Same, only women with all 11 years in these UVs

Woman FE, balanced: **431k women**

Same, only women who appear in 2014 and have all 11 years, in any UV

⇒ **The same four versions, now for women of a fixed age group.**

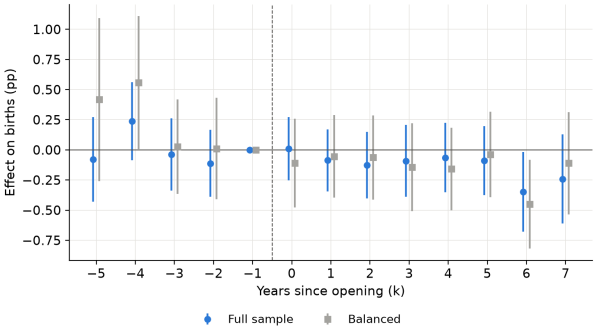
Births fall sharply, in treated and control alike



Vulnerable women aged 18–49.

⇒ **Births per 100 women halve between 2014 and 2024, and treated and control fall together.**

Fertility: no significant effect on births



Measure: share of vulnerable women, 18–35 in 2014, who had a child that year

Start: ≈6 births per 100 women a year, average 2014–2024

Pre-trends: none ($p = 0.25$, balanced $p = 0.27$)

Estimate, first 8 years: **−0.13pp** full, **−0.14pp** balanced, not significant

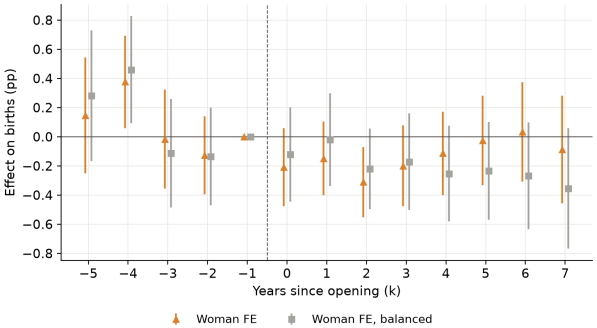
Rules out falls above ≈6% and rises above ≈2%

Last meeting: also no effect, with women 18–49

⇒ **We see no significant effect on births, and can rule out falls above about 6%.**

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$.

Fertility, woman fixed effects



Woman fixed effects: each woman compared with herself, in the first UV where she appears

Start: ≈6 births per 100 women a year

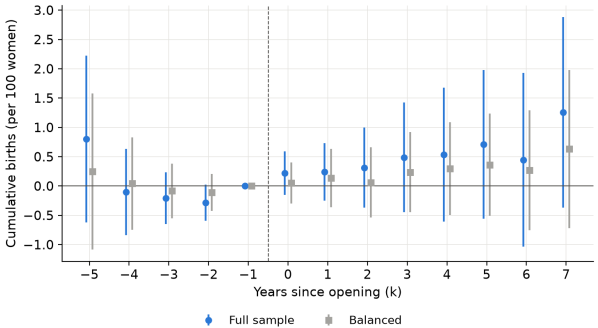
Pre-trends: yes ($p = 0.01$, balanced $p < 0.01$)

Estimate, first 8 years: $-$**0.13pp**, $-$**0.21pp** balanced, not significant

$\Rightarrow$ **With woman fixed effects we see no significant effect either, and both versions have pre-trends.**

Sun–Abraham, 95% CI, ref. $k = -1$.

Fertility: cumulative births



Measure: running count of each woman's births since she first appears in the data

Why: the annual rate shows when women have children; this shows how many

Start: ≈0.38 births per woman

Pre-trends: none ($p = 0.12$, balanced $p = 0.94$)

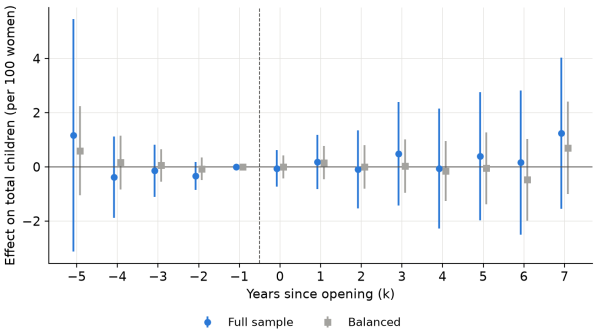
Estimate, first 8 years: **+0.5** full, **+0.3** balanced, per 100 women, not significant

Woman FE: +0.3, with clear pre-trends; parallel trends here require equal birth *rates* (appendix)

⇒ **The number of births does not change significantly: about half a birth more per 100 women.**

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$.

Fertility: total children



Measure: children born before 2014 plus births since she first appears in the data

Why: also counts the children she already had

Start: ≈1.25 children per woman

Pre-trends: none visible, no year stands out

Estimate, first 8 years: **+0.3** full, **+0.03** balanced, per 100 women; no year is significant

⇒ **Counting the children women already had, we also see no significant change.**

Sun–Abraham, weighted, 95% CI clustered by UV, ref. $k = -1$.

Main conclusions so far

Enrollment: rises right when the first preschool opens, from 34% to 37%

Employment: more mothers enter formal work, gradually, from about 55% to 56%. All versions point the same way, at about half the size with woman fixed effects

Earnings including zeros: mothers' average earnings in the UV rise about 3.6%. Woman fixed effects suggest a smaller rise, around 1%

Earnings of mothers with earnings: the UV estimates suggest a rise, but it is not clear when following the same women

Fertility: estimates are small and not significant, suggesting no large change in births

$\Rightarrow$ **Openings bring children into preschool right away and mothers into formal work gradually, with no clear change in births.**

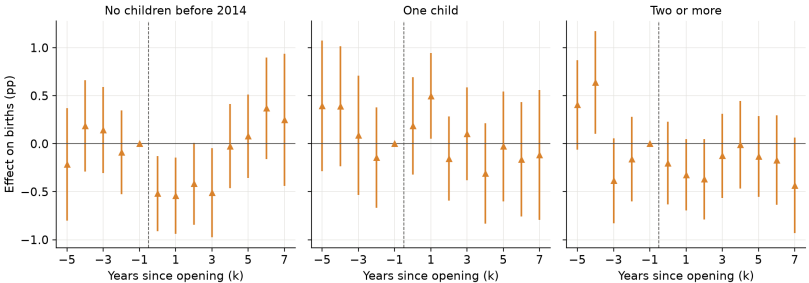
What is pending

Meeting	Topic	Status
Oct 29	Mothers enter at first appearance; balanced women	**Done**
Oct 29	Fertility outcome: cumulative births, total children	**Done**
Oct 29	Fertility outcome: number of births in the UV	**Pending**
Oct 15	Moves in 2022	**Open**: UV fixed from 2022 running
Oct 22	All women, not only mothers (labor)	**Running**
Oct 22	Include UVs with preschools in 2014 (Callaway–Sant'Anna)	**Running**
Oct 15	Effects by child's age at opening	**Running**
Oct 15	Spillovers to neighboring UVs (fertility)	**Running**
Oct 22	Earlier years for pre-trends	**Availability checked**
—	Sensitivity to pre-trends (HonestDiD)	**Next**
Oct 15	First stage for the other 40%	**Pending**
Oct 15	Who fills the new preschool; extended hours	**Pending**
Oct 8	Treatment by initial stock of preschools	**Not started**
Oct 8	Sample by predicted probability of pregnancy	**Not started**

⇒ **Sample changes are done; robustness is running; mechanisms and the 2022 jump come next.**

Appendix

Appendix A1. Fertility by children before 2014 (exploratory)



Woman FE. Each panel is a separate regression: treated vs. control with the same children before 2014; group is fixed

No children before 2014: ≈ -0.5pp the first four years (base 6.1), then back to zero

$\Rightarrow$ **A signal among women without previous children, but it is not repeated at the UV level and starts in the opening year.**

Appendix A2. Why a cumulative outcome has pre-trends more easily

Example with **no effect**: control 6.0 births per 100 women every year, treated 5.8 every year.

k	-4	-3	-2	-1	0	1	2
Gap in the rate	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Coefficient, rate	0	0	0	0	0	0	0
Gap in cumulative births	-0.2	-0.4	-0.6	-0.8	-1.0	-1.2	-1.4
Coefficient, cumulative	$+0.6$	$+0.4$	$+0.2$	0	-0.2	-0.4	-0.6

For the rate, the gap only needs to be stable

For the cumulative outcome, the gap is constant only if both groups have exactly the same rate

If treated had 6.2 instead, the line would rise after the opening with no effect at all

$\Rightarrow$ **A small stable gap in birth rates looks like a pre-trend, or an effect, in cumulative births.**

Appendix A3. Balance in 2014

	Never treated	Treated	Std. difference
Formal employment	55.5%	55.8%	0.03
Monthly earnings, mothers with earnings (2014 pesos)	CLP 296k	CLP 298k	0.02
Months with earnings	8.2	8.1	−0.03
Age	27.9	28.1	0.13
Children under 5	0.68	0.69	0.16
Child density in the UV	0.061	0.064	0.20
Mothers per UV	286	520	**0.50**
UV population	3,198	5,332	**0.44**

Labor sample, weighted by UV population. 299 treated and 3,811 never treated UVs with data in 2014; 2014 UV is the 2016 UV.

⇒ **Mothers look alike in 2014; what differs is UV size.**

Appendix A4. Woman fixed effects: specification

$$E_{it} \;=\; \mu_i \;+\; \lambda_t \;+\; \sum_{k \neq -1} \beta_k \, \mathbf{1}\{\, t - G_{u_0(i)} = k \,\} \;+\; \varepsilon_{it}$$

```
eventstudyinteract empleo_formal d1-d4 d6-d15, cohort(g1_f) control_cohort(nunca_f)
    absorb(rut_inn anio) vce(cluster uv_fija)
```

E_{it}: formal employment of mother i in year t

μ_i: **woman** fixed effect; λ_t: year fixed effect

$u_0(i)$: **fixed UV** = first valid UV where the mother appears, before the opening

$G_{u_0(i)}$: year her fixed UV opens its first preschool → kept even if she moves

Sun–Abraham: cohorts by $G_{u_0(i)}$ (`g1_f`); controls = fixed UV never opens one (`nunca_f`)

$k \leq -5$ binned; ref. $k = -1$; SE clustered by fixed UV

Sample: vulnerable mothers whose fixed UV had no preschool in 2014, first observed before the opening

⇒ **Each mother is compared with herself, treated by the first UV where she appears.**

≈5.6M mother-years.

Appendix A5. More pre-trend years: how far back can we go?

Source	Outcome	From
FPS (women, household)	sample, UV	2007
Preschool enrollment	first stage	2011
Earnings (SII, historical)	labor	2002
Contributions (AFP)	labor	2015
Civil Registry births	fertility	no limit

The catch: UVs before 2016

FPS has **no reliable UV** code

Our crosswalk only reaches back to 2016

Pre-2016 years must be anchored to the woman's **2016 UV** → women who left before 2016 are lost, and earlier moves are invisible

Proposal: start in **2011** → 3 more pre-years for every cohort (the 2015 cohort goes from 1 to 4). Labor 2011–2013: earnings only.

⇒ **Data exist back to 2011 for all outcomes, but UVs before 2016 have to be proxied by the 2016 UV.**

Appendix A6. Full identification assumptions and placebo check

Parallel trends: assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

No anticipation: mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

Exogeneity of timing: opening timing should not correlate with unobserved UV characteristics. **Main threat**: JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

Appendix A7. Data limitation: UV assignment in 2014–2015

FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

Consequence: women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

Idea, not yet tested: fuzzy matching on UV name strings

Appendix A8. Two ways to measure earnings

Example: a UV with 4 mothers. After the opening, **C starts working**, at a somewhat lower wage (CLP million per year).

	A	B	C	D	**Earnings incl. zeros** (all mothers)	**ln earnings** (only mothers who work)
Before	4	4	0	0	$(4+4+0+0)/4 = \mathbf{2.0}$	mean of A, B $= 4$
After	4	4	**3**	0	$(4+4+3+0)/4 = \mathbf{2.75}\uparrow$	mean of A, B, C $\approx 3.6\downarrow$

Earnings including zeros: how much income mothers have *overall* → rises if more mothers work *or* if they earn more

Expressed relative to the year's mean (100): +2 points ≈ +2% for the average mother

ln earnings: how much *working* mothers earn → can fall when new workers enter with lower wages, even if no one's wage falls

⇒ **Earnings including zeros capture new workers; ln earnings only captures wages.**

Appendix A9. Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

What we know

Mixed evidence: positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

Our contribution

First study to use neighborhood-unit (UV) variation: preschools geolocated to the UV and linked to individual administrative records

Fertility as a second margin, alongside labor supply

What is a UV? *Unidad Vecinal*: the official sub-municipal territorial unit — ∼6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

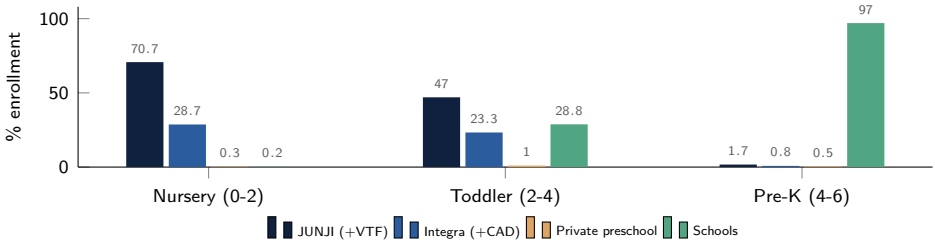
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

Appendix A10. Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

Appendix A11. Identification: staggered-adoption event study

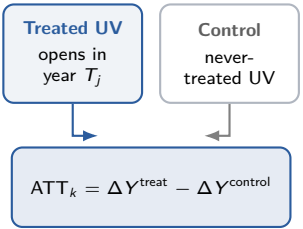
Design: staggered-adoption difference-in-differences with event study

Unit of treatment: the UV, the year it opens its first public preschool

Unit of analysis: **UV** $\times$ **year** panel, built from a woman $\times$ year panel linked to each woman's UV of residence

Fixed effects: **UV + year** (baseline); woman FE on the woman $\times$ year panel as a **robustness check**

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026) — an **intent-to-treat effect of access**, not individual take-up



$\Rightarrow$ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**

Appendix A12. A UV is treated when it opens its first preschool

Treatment definition

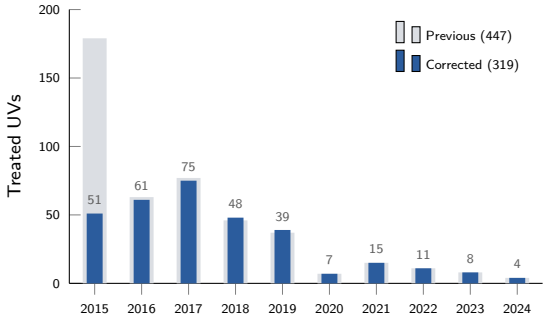
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,394	34.8%
Never-treated	0 in 2014, stays at 0 through 2024	4,174	60.6%
Treated	0 in 2014, opens first center 2015–2024	319	4.6%
Total universe		**6,887**	100%

Source: `base_cobertura_cp.dta`, corrected opening years (Oct 21). Design sample (`muestra_t1==1`) = 4,493 UVs. Previous version: 447 treated, 2,265 excluded.

⇒ **We identify the effect of the first preschool (0 → 1): 319 treated vs. 4,174 never-treated.**

Appendix A13. Treatment cohorts by year



2015 cohort corrected: 179 → **51 UVs** — in 127 UVs the "opening" was a center already operating in 2014 (next slide)

Other cohorts barely change; **2017 is now the largest** (75 UVs)

59% open in 2015–2017; only 45 UVs open after 2019

The 2015 cohort still has **only one pre-period** (2014)

Source: `base_cobertura_cp.dta`, corrected opening years (Oct 21). 319 treated UVs (0 centers in 2014).

⇒ **Openings are spread over 2015–19, no longer concentrated in 2015.**

Appendix A14. Most 2015 "openings" were centers already operating

Problem: JUNJI opening dates come from records that start in March 2015 → centers with no earlier record were dated **01-03-2015** (84% of 2015 openings)

Evidence: older transparency requests give **center** × **year panels** (JUNJI 1996–2023, Integra 2001–2024); JUNJI centers are flat at ∼3,150 a year in 2011–2016

Fix: opening year = first year the center appears in the panels

JUNJI centers "opening" in 2015

Recorded date	Centers	Already there
01-03-2015	444	**427 (96%)**
Other 2015 date	108	15 (14%)

UVs	Before	After
Treated	447	**319**
2015 cohort	179	**51**
Excluded (center in 2014)	2,265	2,394
Never-treated	4,175	4,174

Done: everything re-estimated with the corrected treatment

All results in the main slides use the corrected treatment (319 treated UVs). Previous results are in the appendix.

⇒ **The 2015 cohort was mostly a data artifact: 127 of its 179 UVs already had a center in 2014.**

Appendix A15. Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma\, \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k\, \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

Y_{ut}: average outcome of women in UV u, year t (built from the woman $\times$ year panel). λ_u, λ_t: UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$: average age of mothers in the UV-year (labor market specification).

β_k: effect k years from opening, relative to $k = -1$ (omitted); $k \leq -5$ binned.

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (*JEL*, 2026).

Main threat: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ($k < -1$) test whether trends were already diverging.

$\Rightarrow$ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

Appendix A16. Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV × year)	JUNJI/Integra transparency requests + UV shapefile (MDSF)	Number of centers, treatment status, opening cohort
Women panel (base)	RSH (2016–2024) + FPS (2014–2015)	UV of residence, UV population, child density
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	Annual income, months worked, employment
Fertility	+ Civil Registry birth records (via RIS)	Birth in the year, children under 5
First stage (enrollment)	+ Matrícula parvularia (child-level)	Enrollment, enrollment rate

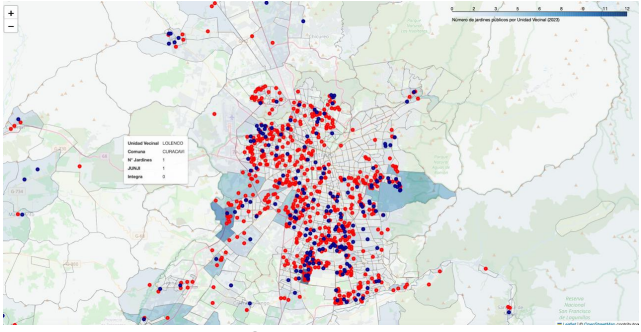
⇒ **All administrative registries, linked at the individual level — no survey self-reports.**

Appendix A17. Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time → we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years



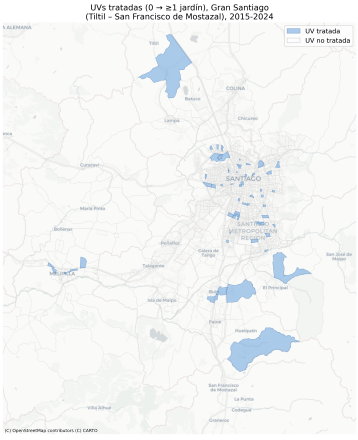
Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: `base_cobertura_cp.dta`.

⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

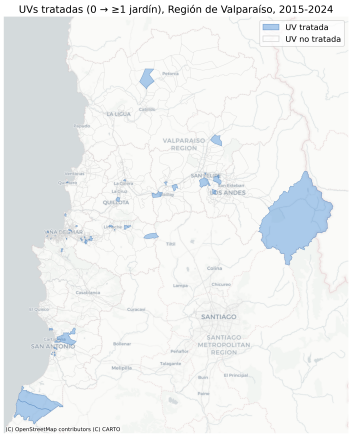
▸ Interactive map 2014–2024

Appendix A18. Examples of treated UVs

Región Metropolitana



Valparaíso



Illustrative examples. Shaded: treated UVs (0 → 1+ preschools), 2015–2024.

Previous results

Slides from the October 15 and 22 meetings, before the data fixes

Who is in each sample, and what we measure

1. Samples (selected at the individual level)

Fertility: women of fertile age (18–49)

Labor market: women with a child <5 *that specific year* — not "ever had a child"

First stage: children of treatment-sample mothers, birth to birth+4

2. Outcomes (averaged at UV × year)

Fertility: share of women who had a child that year (Civil Registry birth records)

Labor market: average annual income, months worked, share employed, dependent vs. independent

First stage: enrollment rate

⇒ **Each outcome is measured on its relevant population, from administrative records, averaged at the UV × year level.**

Treated and never-treated UVs look alike — except in size

Labor market sample (mothers with a child <5) — covariate levels in 2014, before any opening

	Unweighted			Weighted by UV population			
Variable	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	Sig.[a]
Formal employment (share)	0.47	0.48	0.075	0.52	0.51	-0.021	–
Annual income (M CLP)	3.60	3.55	-0.033	3.95	3.89	-0.039	*
Months worked	8.14	8.10	-0.024	8.44	8.37	-0.058	*
Age	29.24	29.25	0.008	29.26	29.29	0.026	
N. children <5	1.11	1.11	0.045	1.11	1.11	0.049	
Hours worked	37.07	37.68	0.099	36.55	36.81	0.068	
Works (share)	0.45	0.45	0.023	0.48	0.48	-0.006	
Education (code)	4.98	5.02	0.028	5.20	5.18	-0.020	
UV population	1126.79	2011.08	**0.410**	3530.92	5287.26	**0.340**	***
Child density 0–5 UV	0.06	0.06	0.155	0.06	0.06	0.139	***
N (UVs)	3,909 / 422			3,909 / 422			

[a] t-test at the woman level (* $p<0.05$, *** $p<0.01$), for reference only; normalized differences prioritized (concern threshold: 0.25); not computed for formal employment (–).

The UV population gap is **expected**: the State prioritizes new preschools where need and population are larger. Same pattern in the fertility sample (App. A4).

⇒ **Selection is in *where* preschools open (UV size), not in *who* lives there → UV fixed effects.**

About 1 in 3 mothers moves across UVs — equally in treated and control UVs

How many move?

33% of mothers change UV at least once while in the sample

7–12% per year

2022: 27% — the year codes were standardized: likely an artifact, not real moves

Do they move toward new preschools?

Group	Moves (% of mother-years)
Treated	12.6
Never-treated	12.3
Excluded	11.9

How we assign a UV to each woman: RSH UV codes are not consistent across years; they are standardized only from 2022. For 2017–2021 we decoded them backwards by following the same women to the next year (majority rule) to place everyone on 2024 boundaries. → **Part of the measured moves may be decoding errors, not real moves** (e.g., the 2022 jump).

Real moves dilute UV-level effects → we also follow each mother from her first UV (woman fixed effects, below).

⇒ **Mothers move often, but not toward UVs that open a preschool.**

Moves measured with 2024 UV boundaries, the same map that defines treatment. Year-to-year changes from 2017 on (2014–2015 UVs are proxied by 2016). 33% = among mothers observed in at least two comparable years.

Results: baseline specification

Results in the next slides use the previous definitions

All women in the RSH — no vulnerability restriction

Labor market: mothers with a child <5 *that specific year*

First stage: children of those mothers; **fertility**: women aged 18–49

UV-level estimates, all treatment cohorts (2015 included)

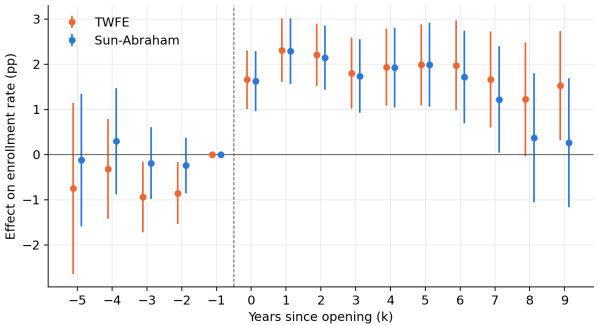
Suggestions from the last meeting

60% most vulnerable, mothers followed over time, individual-level estimates

Incorporated from slide 34 on

⇒ **First the baseline results, then what changes with the new samples.**

First stage: enrollment rises the year a preschool opens and stays higher



Y-axis: change in the enrollment rate (% of children 0–4) in treated vs. never-treated UVs, relative to the year before opening

Treated UVs: **33%** → ≈**35%** (+2pp), from the opening year on

A treated UV has ≈**135 children** aged 0–4 → ≈**3 more enrolled**

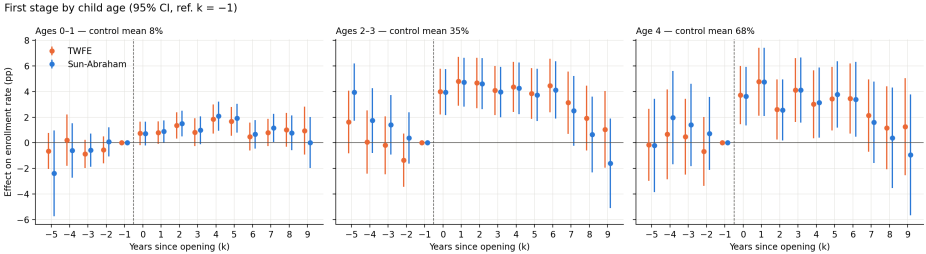
No pre-trends

$k = 7$–9: only early cohorts (mostly 2015) → composition, not a fading effect

⇒ **A new preschool raises enrollment from 33% to ≈35%: ≈3 more children per UV.**

Weighted by number of children. Sun–Abraham and TWFE, 95% CI clustered by UV. Children aged 0–4.

First stage by child age: strongest at ages 2–3



Child age	Control mean	SA $k = 0$–6	Relative
Ages 0–1	8%	+1.3pp	+15%
Ages 2–3	35%	**+4.2pp**	+12%
Age 4	68%	+3.6pp	+5%

Ages 2–3: largest effect, immediate and stable

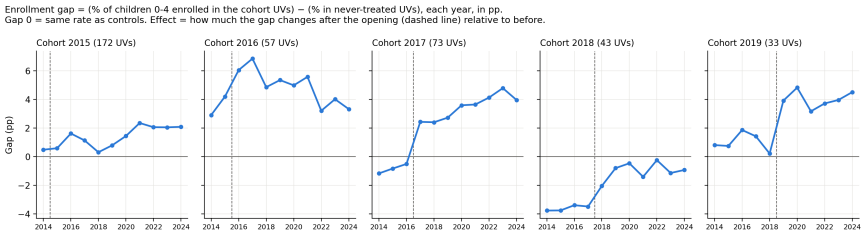
Ages 0–1: small in pp but largest in relative terms; grows over time

Age 4: also positive

Caution: ages are measured at different points in time across datasets (to check)

⇒ **Enrollment rises at all ages, most at ages 2–3.**

An open issue: the 2015 cohort



172 UVs — 41% of treated UVs, ∼half of the sample at $k = 6$, the only cohort at $k = 9$

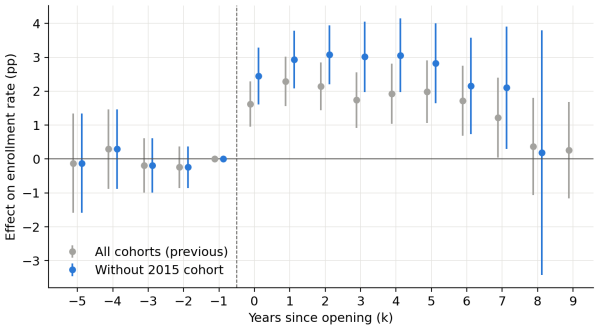
Its effect is **very small** (≈0.7pp in raw gaps) and it has only one pre-period (2014)

Hypothesis: registration error at the edge of the panel — preschools that already operated in 2014 but are recorded as 2015 openings.

Next: carefully review the coverage data provided by JUNJI/Integra.

⇒ **The 2015 cohort pulls the average effect down — possibly a data error to check.**

Without the 2015 cohort, the first stage is ∼50% larger



Same pre-periods (the 2015 cohort has none)

+2.7pp vs. +1.8pp: **≈4 more children** enrolled per UV instead of 3

Jumps at opening and stays stable

Earnings: the effect also roughly doubles without 2015

⇒ The 2015 cohort dilutes the first stage — but it remains small.

All mothers (first stage not yet estimated for the 60% most vulnerable). Average k = 0–7. Sun–Abraham, weighted by number of children, 95% CI clustered by UV.

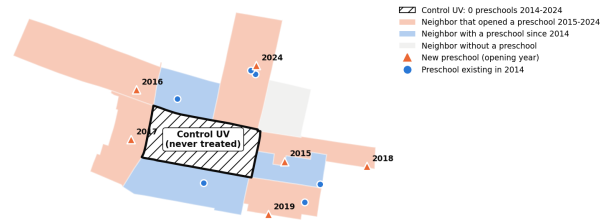
A cleaner control group: never-treated UVs can be exposed too

Concern: children from a never-treated UV can enroll in a new preschool next door → control enrollment rises → **effect underestimated**

We built a **neighbor map**: for every UV, which adjacent UVs had or opened a preschool (shared border, 2024 boundaries)

Example: this control UV has 9 neighbors — **5 open a preschool** during the period

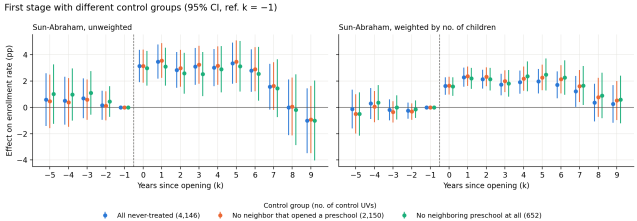
A 'never-treated' UV in La Cisterna
5 of its 9 neighboring UVs open a preschool in the period



⇒ **Goal: a larger first stage, comparing treated UVs with controls that could *not* use a neighbor's new preschool.**

* Neighbors only capture proximity to home. If mothers use preschools near their workplace, spillovers may reach UVs that are not adjacent.

A cleaner control group gives the same first stage



Control group	UVs	Median area	Control mean	SA weighted, $k = 0$–6
All never-treated (main)	4,146	2.8 km^2	33.2%	1.92
No neighbor opened a preschool 2015–24	**2,150**	3.8 km^2	32.7%	**2.14**
No neighboring preschool in any year	652	**6.7 km^2**	28.2%	2.13
Treated UVs	*422*	*1.5 km^2*		

Same effect with all three groups and clean pre-trends. The strictest group is not a good comparison: its UVs are $\sim$**4**$\times$ **larger** than treated ones (isolated areas) $\rightarrow$ preferred clean control: **no neighbor opened**.

$\Rightarrow$ **Spillovers to controls do not explain the small first stage.**

Why is the first stage small?

What we checked

2015 cohort: very small effect, possibly a data error → without it, ≈3pp

Cleaner control group: same ≈2pp → spillovers to controls are not the reason

What our outcome measures

Whether children **living** in the treated UV are enrolled in **any** preschool

A new preschool can fill up without moving our outcome

It raises enrollment only if it takes local children who were **not enrolled before**

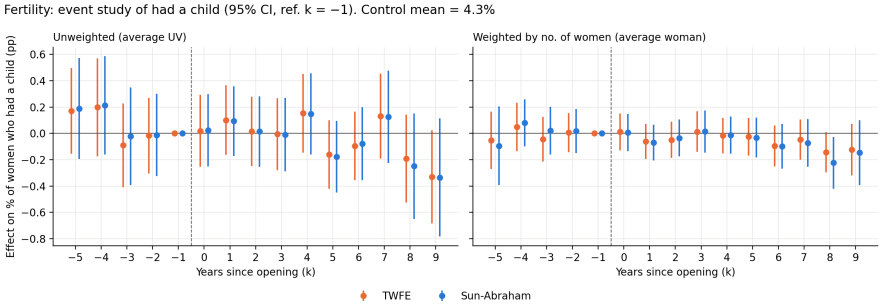
Substitution: local children already enrolled elsewhere switch to the new preschool → no net change

Mothers use other preschools, e.g. near their workplace in another UV → new places go to children from other UVs

Future work: enrollment records show which preschool each child attends → who fills the new preschool, where they live, and which preschools local mothers use.

⇒ **Enrollment of local children can move little, even if the new preschool fills up.**

Fertility: no effect yet



Treated and control UVs decline together; **no jump at opening**

Clean pre-trends

A **fairly precise zero**: rules out large effects

Still checking: cumulative outcome (having had a child)

⇒ **Expected: having a child depends on many factors, and a nearby preschool might weigh little.**

Women aged 18–49. Outcome: share who had a child in the year (control mean 4.3%). Controls for average age.

Mothers' formal employment rises by 1pp from the fourth year



Nothing at opening; **significant from** $k = 4$; no pre-trends ($p = 0.82$)

Why gradual? Two possible reasons

Who is in the sample: if the child is already 3–4 at opening, the mother barely uses it; later, more mothers had it since birth

Returning to work takes time

How to tell them apart:

Keep mothers until 2024 → no one leaves when her child turns 5

Woman FE → each mother compared with herself

If still gradual → it is the time to return to work

⇒ A new preschool raises mothers' formal employment by ≈1pp from the fourth year.

Weighted by number of mothers. Mothers with a child <5 that year. Employed = any taxable income or contribution.

Changes to the samples

1. Only the 60% most vulnerable (labor and fertility samples)

The population **eligible** for JUNJI/Integra preschools → removes women who could not be treated

Vulnerable = RSH bracket ≤60 (FPS score ≤13,500 in 2014–2015, new samples only)

Fixed per woman: her classification in the **first year she appears**, kept for all years → does not respond to the opening

2. Mothers stay in the labor sample (in progress)

Before: a mother is in the sample only **while** she has a child <5

Now: she enters the first year she has a child <5 and **stays until 2024**, even after the child turns 5 → the sample no longer changes with births and children aging out

⇒ **We now focus on eligible mothers and follow the same mothers over time.**

Results below: change 1 with RSH only and the previous mother definition. FPS and change 2: new samples being built.

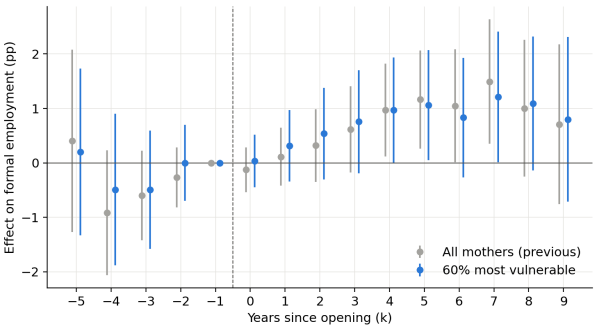
Where each change stands

	First stage	Labor market	Fertility
60% most vulnerable	pending	**done** (slides 36–38)	pending
Mothers followed until 2024	—	running	—
Woman FE, fixed UV	—	**preliminary** (slide 39)	—
Without the 2015 cohort	**done** (slide 28)	**done**: similar	pending
Earnings including zeros	—	**done** (slide 38)	—

Labor results for vulnerable mothers use RSH only and the previous mother definition. The first stage for their children comes next.

⇒ **Labor results already reflect the vulnerable sample; first stage and fertility come next.**

Employment: same effect among the 60% most vulnerable



No pre-trends (joint test $p = 0.79$)

Gradual, $\approx$+1pp from the 4th year; average **+0.8pp** on a 46% base ($\approx$+1.6%), significant

Why so similar to all mothers? If the non-vulnerable third had no effect, we would expect $\approx$1.1pp (same effect, fewer mothers). But:

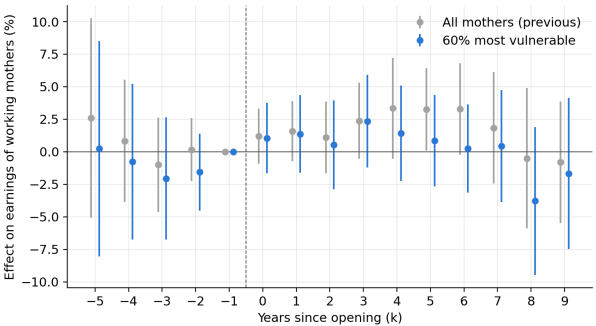
- Those excluded are RSH middle class, not rich; vulnerable mothers are 2/3 of the sample
- Eligibility is not strict: free places go to other families

Next: estimate the non-vulnerable separately.

$\Rightarrow$ **The employment effect holds in the eligible population.**

Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV, ref. $k = -1$.

Working mothers do not earn more



In earnings: **only mothers who work**

No pre-trends; after opening, ≈**0** among vulnerable mothers (+0.3%, not significant)

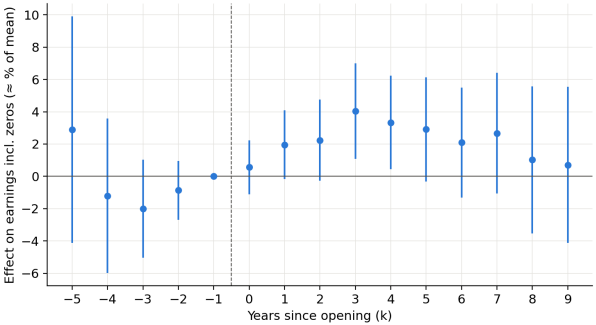
All mothers: slightly higher (≈+3% at $k = 3$–6), also not significant

Months worked: also no effect

⇒ **The earnings gain does not come from higher wages.**

Sun–Abraham, weighted by number of mothers with earnings, 95% CI clustered by UV, reference $k = -1$. Mothers with a child <5 that year.

Earnings rise because more mothers work, not because they earn more



Earnings **including zeros**: average over *all* mothers, those without earnings count as 0

Y-axis: 100 = average mother that year → **+2 points ≈ +2%**

Average after opening: ≈**+2%**, peak at $k = 3$–4 (significant at 10%)

∼**3/4** of it is explained by more mothers working

Among mothers who work: earnings and months worked **do not change**

⇒ **The preschool changes *whether* mothers work, not how much those already working earn.**

60% most vulnerable. Earnings relative to the year's mean (100). Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV.

Woman fixed effects: a design to discuss

Approach

Compare **the same mother** before and after the opening → removes her fixed traits (e.g., attachment to work)

Treatment from her **UV before the opening**, kept if she moves (intention-to-treat)

Movers are **not dropped**: staying could itself respond to the preschool

Results so far (preliminary)

Employment: no pre-trends; gradual, significant from the 4th year, $\approx +1.6$pp by $k = 9$ — same pattern as UV level

Limitation

Does **not** address *where* and *when* preschools open — that happens at the UV level

Only helps if who lives in treated UVs changes after the opening

Open questions

Is woman FE worth it? Moves are similar in treated and control UVs (12.6% vs. 12.3%) → it may mainly serve as a robustness check

Fixed pre-opening UV or contemporaneous UV?

Should movers be kept or dropped?

⇒ **Same employment result following the same mothers — input welcome on the design.**

Women first observed in 2014–2015: UV proxied by 2016 (App. A7). To be re-estimated with the new mother sample.

Running and pending

Running

New samples: vulnerable mothers followed until 2024; fertility and first stage for the 60% most vulnerable

Woman fixed effects with the new sample: average effect and SE

Pending

Re-run everything with the corrected treatment (319 treated UVs): samples, balance, first stage, employment, earnings, fertility

First stage among the 60% most vulnerable

All outcomes with the new samples

Who fills the new preschool? Enrollment in JUNJI/Integra vs. other providers

Balanced event study (same cohorts pre and post)

The other 40% separately: is the effect concentrated among vulnerable mothers?

Treatment by initial stock of preschools

Sample by predicted probability of pregnancy